

SURV 621/SURVMETH 621 Fundamentals of Data Collection

In-class exercise: Applying the Total Survey Error framework

Monday, August 26, 2024

Using the Total Survey Error model, critique the estimates likely to be obtained from the survey described below. Indicate any concerns you may have regarding any or all of the following:

- Validity of chosen measurement
- Measurement error
- Processing error
- Coverage error
- Sampling error
- Nonresponse error
- Post-survey adjustment error

You are designing a survey for a client who is interested in learning about the distribution of household spending across different categories of expenditure for consumers in a particular metropolitan area. The broad categories of interest include food and beverages, apparel, recreation and entertainment, medical care, education, transportation, communication, housing, and other goods and services.

To collect this information, you propose a random digit dialing (RDD) survey. Interviewers will dial all landline phone numbers in randomly selected blocks of 100 numbers associated with the metropolitan area that contain at least one listed residential phone number. If an adult age 18 or older answers the phone and confirms that the number dialed is a residential number, that person will be asked to complete an interview; if a person under age 18 answers the phone, the interviewer will ask to speak to an adult.

Interview subjects will be asked to report all of the places where they shopped over the last 7 days; the items they bought; and how much they spent on each of those items. Information on subjects' age and sex also will be collected. Interviews will continue until 2,500 responses have been obtained.

Once data collection has concluded, purchased items will be coded into expenditure categories and each subject's total reported spending by expenditure category will be calculated. The responses will be weighted so that the age/sex composition of the sample matches the age/sex composition of the population in the metropolitan area, with the responses from age/sex groups that are under-represented weighted more heavily and the responses from age/sex groups that are over-represented weighted less heavily. Total spending by expenditure category based on the weighted responses will be used to estimate the distribution of household spending across categories for consumers in the metropolitan area.